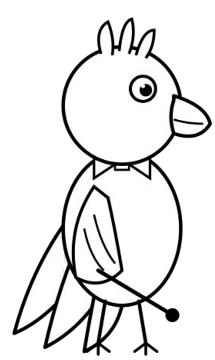
Grade 3
Ontario Curriculum
Algebra (Patterns and Relationships)

The Pattern Parade — Operations, Big Numbers, and Justifying the Rule



Unit Overview: The Pattern Parade — Operations, Big Numbers, and Justifying the Rule

Grade: Grade 3 | Strand: Algebra (Patterns and Relationships)

Target Expectations: C1.1, C1.2, C1.3, C1.4

Learning Goal: Children will identify the operation in repeating, growing, and shrinking patterns; translate patterns across shapes, numbers, and tables of values; determine compound and multiplicative rules; and justify predictions for far terms using whole numbers up to 1000.

Lesson	Lesson Title	Learning Goal
Day 1	**Welcome Back, Pattern Conductors**	I can identify the operation in a pattern and describe whether it is repeating, growing, or shrinking.
Day 2	**Multi-Step Pattern Rules**	I can name a multi-step pattern rule and use it to build the next term.
Day 3	**Tables of Values for Big Numbers**	I can build a table of values for a number pattern up to 1000 and use it to predict a faraway term.
Day 4	**Justify and Generalize**	I can predict term 50 of a pattern and justify my prediction using the rule and the term number.
Day 5	**Lead the Number Parade to 1000**	I can build a number parade up to 1000, organize it as a table of values, predict term 50, and justify my rule.

Lesson 1: Welcome Back, Pattern Conductors

Goal: Patterns can repeat, grow, or shrink — and the rule can use addition, subtraction, multiplication, or a combination.
Expectation: C1.1

Learning Goal: "I can identify the operation in a pattern and describe whether it is repeating, growing, or shrinking."
Materials: M1_animal_cards, M2_parade_strip, M3_sound_movement_cards, M6_anchor_chart_words

1. Minds On (12 mins): Coco's Five Mystery Parades

Coco is back, parade-leaders. This year he brought FIVE different parades — and one of them is a brand-new kind we have never seen before. Can we name the operation that drives each one?

Bring children to the carpet. Hold up Coco. Say to students: 'Welcome back, parade-leaders. In Grade 2 you sorted patterns into three types — repeating, growing, shrinking — and…

2. Action (28 mins): The Operation Hunt

• Whole-group sort and label: lay out four MORE pre-built parades and four label cards (REPEATING, GROWING +N, GROWING ×N, SHRINKING -N).
• Partner build round 1: each pair gets two sets of Animal Cards (M1) and one Parade Strip (M2).
• Partner build round 2: switch to a SHRINKING parade.
• Real-life operation hunt (8 min): pairs walk the classroom (and hallway if safe) with a clipboard.

3. Consolidation (10 mins): Operation Showcase

Discuss:
• What was the most surprising operation you saw or built today?
• Did anyone find a real-life pattern that uses MULTIPLICATION or DIVISION as the operation?
• If a parade keeps doubling (×2 each term), how big does it get after 10 terms?
• Coco is asking — are operations the same thing as rules?

Exit: On the way out the door, each child gives the teacher one example using the…

Lesson 2: Multi-Step Pattern Rules

Goal: A pattern rule can be one step (+5) or a multi-step recipe (+3 then -1) — as long as the recipe stays the same.
Expectation: C1.3

Learning Goal: "I can name a multi-step pattern rule and use it to build the next term."
Materials: M1_animal_cards, M2_parade_strip, M5_core_loop

1. Minds On (12 mins): Coco's Recipe Reveal

Yesterday Coco showed us five parades, and one had a SECRET COMPOUND RULE we did not crack. Today we crack it open — and we discover that the rule of a pattern can be a TWO-STEP RECIPE.

Bring children to the carpet. Lay parade 1 (single-step: 5, 10, 15, 20, 25) on the…

2. Action (28 mins): Build the Recipe, Test the Rule

• Whole-group demo of a multi-step build: take a fresh number parade, start at 2, apply the rule '+3 then -1' for 6 terms.
• Partner build round 1 — single-step parades: each pair gets Animal Cards (M1), one Parade Strip (M2), and a stack of sticky notes.
• Partner build round 2 — multi-step parades: now each pair builds ONE compound number parade.
• Stretch challenge for fast pairs: invent a 3-step compound rule (e.g., +5, then +3, then -2 — cycle of 3) and build at least 9 terms.

3. Consolidation (10 mins): Recipe Showcase

Discuss:
• Who built the most surprising multi-step rule today?
• When you used a +N then -M rule, what was the NET effect after one full cycle?
• Can a compound rule make a parade SHRINK overall, even though one of its operations is +N?
• Coco wants to know — what is the difference between a compound rule like '+3 then -1' and just describing the pattern term by…

Lesson 3: Tables of Values for Big Numbers

Goal: A table of values organizes any pattern, and we need it most when numbers get big enough that we cannot draw every term.
Expectation: C1.2

Learning Goal: "I can build a table of values for a number pattern up to 1000 and use it to predict a faraway term."
Materials: M1_animal_cards, M2_parade_strip, M5_core_loop

1. Minds On (12 mins): Coco's Three Faces, Big Numbers Edition

Coco brought ONE pattern today and put on THREE costumes — shapes, numbers, and a TABLE. Today the numbers go big. By the end of class, we will predict term 20 of a pattern without drawing 20 cells.

Bring children to the carpet. Lay all three representations of the +25 pattern in a...

2. Action (28 mins): Translate Across Three, Predict the Faraway Term

- Whole-class table-building demo: choose a fresh growing rule WITH the class (e.g., +50 starting at 50).
- Partner translation round 1: each pair gets Animal Cards (M1), one Parade Strip (M2), one Core Loop Frame (M5), and a piece of paper for the table.
- Partner translation round 2: pairs swap rules with another pair.
- See the expanded walkthrough below for the keystone reveal moment when pairs realize the table not only ORGANIZES the pattern but also REVEALS the relationship between term...
- Stretch challenge: pairs build a table of values for a multi-step rule from Lesson 2...

3. Consolidation (10 mins): Far-Term Showcase

Discuss:

- Who predicted the BIGGEST term value today?
- Look at a +N rule starting at N.
- When IS a table of values most useful — small numbers or big numbers?
- Coco wants to know — can we predict term 100 of a +5 rule starting at 5?

Lesson 4: Justify and Generalize

Goal: A Pattern Conductor doesn't just predict — they JUSTIFY using the rule, the start, and the number of jumps.
Expectation: C1.3

Learning Goal: "I can predict term 50 of a pattern and justify my prediction using the rule and the term number."
Materials: M1_animal_cards, M2_parade_strip, M4_missing_card, M5_core_loop

1. Minds On (12 mins): Coco's Far-Term Mystery

Coco wrote three patterns on the board with HUGE gaps — term 5, term 7, term 20. We don't have time to draw every cell. Today we use the rule to JUMP.

Bring children to the carpet. Lay the three setup patterns in a line. Hold up Coco.

2. Action (28 mins): Detectives at Work — Predict Far, Justify Always

• Whole-class far-prediction model: take a fresh growing pattern (rule +6 starting at 6).
• Partner round 1 — single-step far prediction.
• Partner round 2 — multiplicative far prediction.
• Partner round 3 — backward justification: pairs build a pattern with one VISIBLE term in the middle (e.g., term 5 = 32) and a known rule (×2).
• Stretch challenge — compound rule far prediction.
• Whole-class debrief AND circulate-track: bring 2-3 pairs back to share justifications aloud.

3. Consolidation (10 mins): Justify and Generalize Showcase

Discuss:
• Who computed the BIGGEST term value today?
• When did you use multiplication as a SHORTCUT instead of skip-counting?
• If your rule is ×3 starting at 3, at what term does the value cross 1000?
• Coco wants to know — what is the difference between PREDICTING and JUSTIFYING?
• What is one GENERALIZATION you discovered today that we can write on the class chart?

Exit: On the way out the door, each child says one full justification sentence using the frame: 'I started at ___, the rule is ___, I jumped ___ times, so term ___ is ___.' Teacher records who can produce…

Lesson 5: Lead the Number Parade to 1000

Goal: Pattern rules also work with numbers up to 1000 — and we now plan, build, translate, table, and lead a parade ourselves.
Expectation: C1.4

Learning Goal: "I can build a number parade up to 1000, organize it as a table of values, predict term 50, and justify my rule."
Materials: M1_animal_cards, M2_parade_strip, M3_sound_movement_cards, M4_missing_card

1. Minds On (12 mins): From Animals to Numbers up to 1000

All week we have practiced the moves of a Pattern Conductor: identify the operation, name the rule, build the table, predict the faraway term, justify formally. Today YOU put it all together — and Coco's parade goes all the way up to 1000.

Bring children to the carpet. Hold up Coco. Beside him, lay the +75 example (75, 150…

2. Action (38 mins): Plan, Build, Translate, Table, Predict-and-Justify, Lead

- Phase 1 — PLAN (5 min).
- Phase 2 — BUILD an animal parade (6 min).
- Phase 3 — TRANSLATE to a number parade up to 1000 (6 min).
- Phase 4 — TABLE (6 min).
- Phase 5 — PREDICT AND JUSTIFY term 50 (6 min).

3. Consolidation (10 mins): Junior Pattern Conductor Ceremony — Grade 3 Edition

Discuss:
- Who would like to LEAD the WHOLE class through their parade?
- Whose rule produced the BIGGEST term-50 value?
- What was the trickiest part of the summative?
- What is one GENERALIZATION you discovered this week that you will remember for Grade 4?
- Coco wants to know — are you a Pattern Conductor now?

Exit: Each child receives a Junior Pattern Conductor certificate (the teacher reads it…

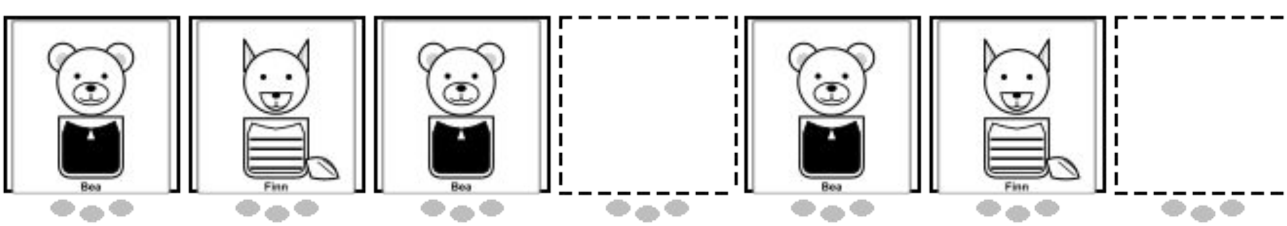
Name: ____________________________ Date: _________________________

Spot the Operation

Learning Goal: I can identify the operation in a pattern and describe whether it is repeating, growing, or shrinking.

Instructions for the Student:

1. Look at each pattern.
2. Look at this growing parade.
3. Look at each real-life thumbnail.



(3 parts on the printable worksheet)

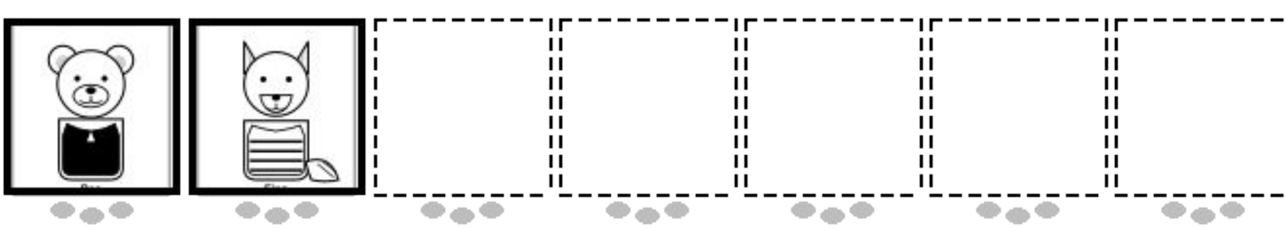
Name: ______________________ Date: ______________________

Two-Step Recipes

Learning Goal: I can name a multi-step pattern rule and use it to build the next term.

Instructions for the Student:

1. Look at each pattern.
2. The rule is +4 then -1.
3. Pick your own starting number AND a 2-step compound rule.



(3 parts on the printable worksheet)

Name: ____________________ Date: ____________________

Tables Tell the Future

Learning Goal: I can build a table of values for a number pattern up to 1000 and use it to predict a faraway term.

Mission

1. The rule is +30 starting at 30.
2. Fill in the missing value AND write the rule.
3. Pick your own +N rule with N between 25 and 100.

Worksheet area

WS03 P1 PARADE

(3 parts on the printable worksheet)

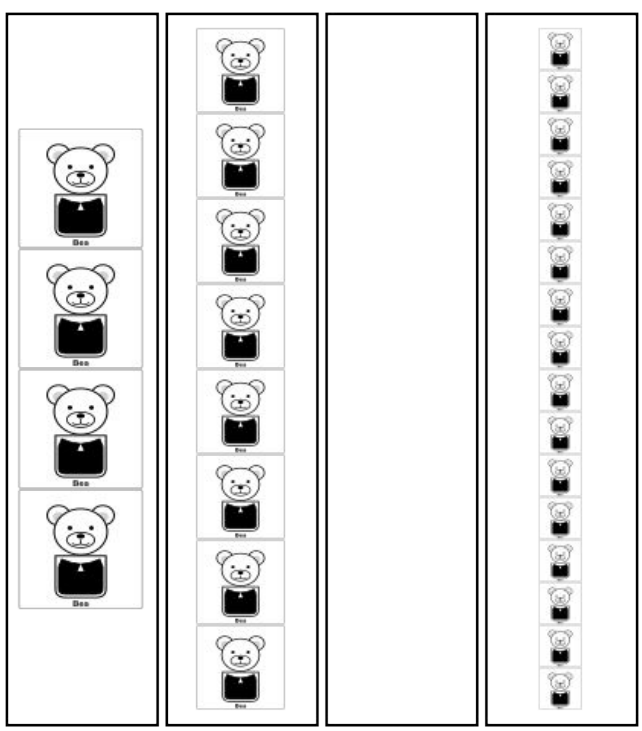
Name: ______________________________ Date: ___________________________

Justify the Faraway Term

Learning Goal: I can predict term 50 of a pattern and justify my prediction using the rule and the term number.

Mission

1. For each pattern, write the rule, predict term 10, and justify using the sentence frame.
2. Look at the table.
3. Coco's pattern has rule ×2.



(3 parts on the printable worksheet)

Name: ______________________ Date: ______________________

Lead the Number Parade to 1000

Learning Goal: I can build a number parade up to 1000, organize it as a table of values, predict term 50, and justify my rule.

Instructions for the Student:

1. You are the Conductor today.
2. Use your rule to fill all 8 cells of the animal parade.
3. Fill in the Value column with your 8 number-parade values.

0–100 chart

0	1	2	3	4	5	6	7	8	9
10	11	12	13	14	15	16	17	18	19
20	21	22	23	24	25	26	27	28	29
30	31	32	33	34	35	36	37	38	39
40	41	42	43	44	45	46	47	48	49
50	51	52	53	54	55	56	57	58	59
60	61	62	63	64	65	66	67	68	69
70	71	72	73	74	75	76	77	78	79
80	81	82	83	84	85	86	87	88	89
90	91	92	93	94	95	96	97	98	99
100									

(3 parts on the printable worksheet)

Parade Animal Cards

(M1_animal_cards)

Reusable physical cards used by pairs to build, manipulate, read, translate, and fix parade patterns.

PRINT SPECS
Size: letter portrait
Colour: bw
Pages per set: 2
Laminate: yes

USE IN CLASS
Quantity: 2 sets (32 cards) per pair
Lessons: 1, 2, 3, 4, 5
Prep: ~30 min

TEACHER PREP
1. Print 2 sets per pair for Grade 3 (one set is 16 cards = ~10 sets total for a.
2. Optional but strongly recommended.
3. Cut along the thin solid lines into 16 individual 3in × 3in cards per set.
4. Store each pair's 32-card set in a labelled zip-top bag.
5. Grade 3 storage tip.
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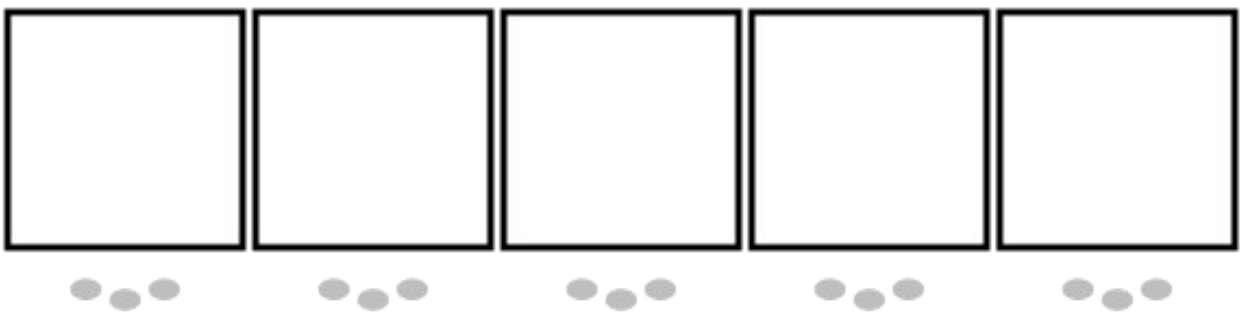
Blank Parade Strip

(M2_parade_strip)

A printable strip with 5 cells.



PRINT SPECS
Size: letter landscape
Colour: bw
Pages per set: 1
Laminate: yes

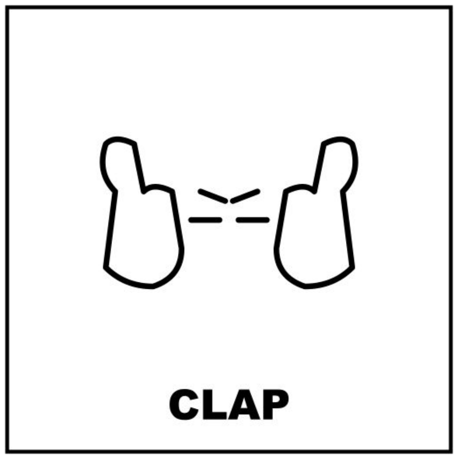
USE IN CLASS
Quantity: 1 strip per pair plus 2 spare
Lessons: 1, 2, 3, 4, 5
Prep: ~15 min

TEACHER PREP
1. Print enough pages to yield ~12 strips (1 per pair plus 2 spare = 6 pages.
2. Laminate the pages.
3. Cut along the horizontal cut line between the two strips.
4. Distribute one strip per pair; keep 2 extras at the teacher table.
5. Bring a class set of dry-erase markers and small cloths/erasers.

Sound and Movement Cards

(M3_sound_movement_cards)

An 8-card set used by the class.



PRINT SPECS
Size: letter portrait
Colour: bw
Pages per set: 1
Laminate: yes

USE IN CLASS
Quantity: 1 teacher set (8 cards)
Lessons: 1, 5
Prep: ~10 min

TEACHER PREP
1. Print 1 page (Grade 3 needs only the teacher's whole-group set).
2. Laminate for durability.
3. Cut along the cut-lines into 8 individual cards.
4. Store in a labelled envelope.
5. Optional colour suggestion: B&W works well; colour for variety.

Missing Mystery Card

(M4_missing_card)

A small card showing a question mark inside a dotted outline.

PRINT SPECS
Size: letter portrait
Colour: bw
Pages per set: 1
Laminate: yes

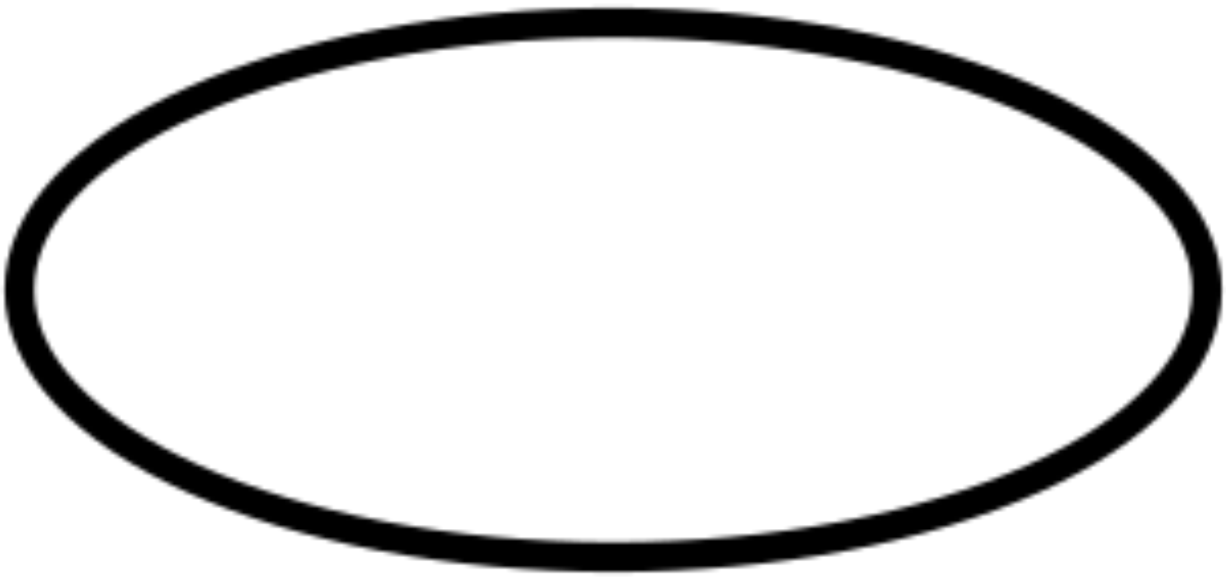
USE IN CLASS
Quantity: 5 cards per pair
Lessons: 4, 5
Prep: ~15 min

TEACHER PREP
1. Print 5 pages (60 cards = 5 cards per pair × 10 pairs + spares).
2. Laminate before cutting.
3. Cut into individual 2in × 2in cards.
4. Distribute 5 cards per pair before Lesson 4.

Core Loop Frame

(M5_core_loop)

A reusable physical frame.



PRINT SPECS

Size: letter portrait
Colour: bw
Pages per set: 1
Laminate: yes

USE IN CLASS

Quantity: 1 frame per pair plus 2 spares
Lessons: 2, 3, 4, 5
Prep: ~15 min

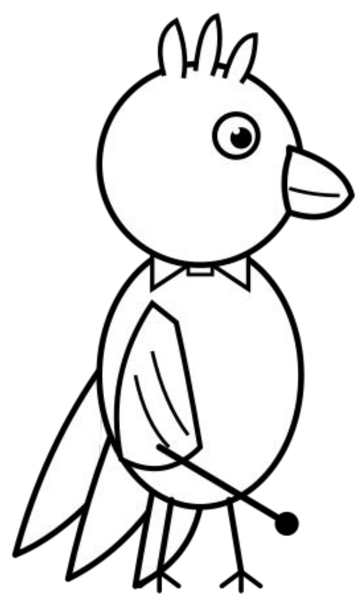
TEACHER PREP

1. Print on cardstock if available — durability matters.
2. Alternative: print on regular paper and laminate.
3. Cut around the outside of each oval to produce 4 frames per page (3 pages = 12 frames).
4. Distribute one frame per pair before Lesson 2.
5. Provide sticky notes per pair for writing rule labels (single-step OR multi-step).

Coco the Conductor Puppet

(char_coco_puppet)

A printable Coco illustration for the teacher's popsicle-stick puppet.



PRINT SPECS
Size: letter portrait
Colour: bw
Pages per set: 1
Laminate: yes

USE IN CLASS
Quantity: 1 puppet for the teacher
Lessons: 1, 2, 3, 4, 5
Prep: ~10 min

TEACHER PREP
1. Print 1 page on cardstock (or regular paper + laminate).
2. Cut around Coco's silhouette including the bottom tab.
3. Glue or tape a popsicle stick behind the tab.
4. Optional: laminate before cutting.
5. Store flat in a folder.
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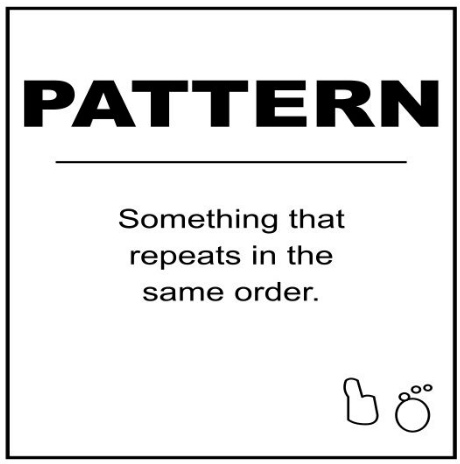
Anchor Chart Word Cards

(M6_anchor_chart_words)

A printable set of four word cards (PATTERN, CORE, MISSING, EXTEND).



PRINT SPECS
  Size: letter portrait
  Colour: bw
  Pages per set: 1
  Laminate: yes

USE IN CLASS
  Quantity: 1 set per class
  Lessons: 1, 2, 4
  Prep: ~12 min

TEACHER PREP
1. Print 1 page (1 set per class).
2. Laminate for durability before cutting.
3. Cut along the cut-lines into 4 individual word cards.
4. On Day 1, after introducing 'pattern', tape PATTERN to the chart.
5. On Day 2, tape CORE.

…

Name: ______________________ Date: ______________________

Summative Assessment Rubric

Expectation	Level 1 (Beginning)	Level 2 (Developing)	Level 3 (Achieving)	Level 4 (Exceeding)
C1.1 Identify repeating elements and operations in patterns	Names what's next only with prompts.	Predicts most of the time. Creates 2-element pattern with help.	Confidently predicts and creates 2- or 3-element patterns.	Creates 3- or 4-element cores; spots patterns across reps.
C1.2 Translate patterns; build tables of values	Treats different reps as different patterns (surface only).	Produces one 2nd rep with help; structure may drift.	Independently translates a rule into 2 different reps.	Translates into 3+ reps including invented ones.
C1.3 Determine rules (incl. growing/shrinkin g +/-); justify	Cannot consistently name a rule or fill missing.	Names a rule with prompts; fills missing at end only.	Names rule independently; fills missing anywhere; justifies.	Names invented rules; extends both directions independently.
C1.4 Use patterns to describe whole numbers up to 1000	Does not yet translate animals to numbers.	Builds 2-element number parade with small numbers, with help.	Builds repeating number parade; explains relationship.	Uses larger numbers; predicts cells far beyond.

Junior Pattern Conductor Certificate

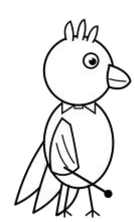
This certificate is proudly presented to:

____

For successfully completing the 5-Day Pattern Parade unit, identifying operations and multi-step rules, building tables of values, predicting faraway terms with formal justification, and leading a number parade up to 1000. A pattern parade certificate. Coco the Conductor congratulates the student.

By completing this unit, this student has demonstrated:

- Identifying operations in repeating, growing, and shrinking patterns — including multiplication and compound rules
- Naming pattern rules using formal notation (+N, -N, ×N, and multi-step recipes)
- Translating patterns across shapes, numbers, AND tables of values



Marketplace Listing

A 5-day Ontario Grade 3 unit on Algebra (C1.1–C1.4), framed as a parade led by Coco the Conductor and four Canadian animals.

Price: $15.99 CAD
Pages: ~46
Classroom time: 255 min
Teacher prep: ~110 min

Top tags: patterns, grade 3, Ontario, Ontario curriculum, math, mathematics, algebra, patterns and relationships

What's included:

- 5 fully-scripted lesson plans (50–55 minutes each, ~255 minutes total of teaching time)
- 5 student worksheets aligned to each lesson, with name/date fields, child-voice learning goals, and Grade 3 sentence frames for formal justification
- 1 mid-unit formative assessment ('Tables and Operations Check') administered between Lesson 3 and Lesson 4
- 1 end-of-unit reflection sheet with 5 prompts for student self-assessment, including a yes/no audit of all four Grade 3 expectations including the new ones
- 7 printable teacher manipulatives: parade animal cards (2 sets per pair to support large-value patterns), parade strips (laminated for dry-erase number…
- 4 class observation trackers: 1 diagnostic (Lesson 1) plus 3 formative (Lessons 2, 3, 4) — each with Grade 3-specific column headers including operations,…
- 1 four-level summative rubric covering all four expectations (C1.1–C1.4) with Grade 3 descriptors aligned to the 0–1000 number range, table-of-values…
- 1 summative administration guide tying the worksheet, lesson, and rubric together for the teacher
- Junior Pattern Conductor certificate (one per child, Grade 3 edition)
- Detailed expanded walkthrough for Lesson 3 (the keystone — table of values reveals the multiplication relationship), including expected student misconceptions…
- Differentiation tracks built into every lesson: emerging learners, extending learners, ELL/IEP, and movement-first learners
- Real-life pattern hunt (Lesson 1) with Canadian and multiplicative contexts: calendar, hockey scoreboard, paper-fold doubling, ice-cube melting